

$$T(rn) = \frac{1}{r} (T(n) + M(n)) \quad (1-الف)$$

$$h = \frac{b-a}{rn} \quad x_0 = a, x_1, x_2, \dots, x_{rn} = b$$

$$x_i = a + ih$$

$$\Rightarrow T(n) = \sum_{i=0}^{n-1} \frac{r h}{r} (f(x_{ri}) + f(x_{ri+r})) = h (f(x_0) + r \sum_{i=1}^{n-1} f(x_{ri}) + f(x_{rn})) \quad (1)$$

$$\Rightarrow T(rn) = \sum_{i=0}^{rn-1} \frac{h}{r} (f(x_i) + f(x_{i+1})) = \frac{h}{r} (f(x_0) + r \sum_{i=1}^{rn-1} f(x_i) + f(x_{rn})) \quad (2)$$

$$\Rightarrow M(n) = \sum_{i=0}^{n-1} r h f(x_{ri+1}) \quad (3)$$

$$\xrightarrow{(1), (2), (3)} \frac{1}{r} (T(n) + M(n)) = \frac{h}{r} (f(x_0) + r \sum_{i=0}^{n-1} f(x_{ri+1}) + r \sum_{i=1}^{n-1} f(x_{ri}) + f(x_{rn}))$$

$$= \frac{h}{r} (f(x_0) + r \sum_{i=1}^{rn-1} f(x_i) + f(x_{rn})) = T(rn)$$

$$M \in [a, b] \text{ ولي: } E_{T(rn)} = \frac{b-a}{1r} \times \left(\frac{h}{r}\right)^2 \times f''(M) = \frac{T(rn) - T(n)}{r}$$

$$\int_{x_i}^{x_{i+1}} f(x) dx = \frac{h}{r} (f(x_i) + f(x_{i+1})) - \frac{h^2}{1r} f''(M)$$

طبق روش تفاضل دارم:
 $\overline{h} = \frac{b-a}{rn}$

$$\Rightarrow E_{T(rn)} = \int_{x_0}^{x_{rn}} f(x) dx - T(rn) = \sum_{i=0}^{rn-1} \left(\frac{h}{r} (f(x_i) + f(x_{i+1})) - \frac{h^2}{1r} f''(M_i) \right)$$

$$= \sum_{i=0}^{rn-1} \frac{h}{r} (f(x_i) + f(x_{i+1})) - \sum_{i=0}^{rn-1} \frac{h^2}{1r} f''(M_i) =$$

$$\Rightarrow E_{T(rn)} = -\frac{h^2}{1r} \sum_{i=0}^{rn-1} f''(M_i) \quad (x_i \leq M_i \leq x_{i+1})$$

$$\Rightarrow E_{T(rn)} = -\frac{h^2}{1r} \sum_{i=0}^{rn-1} f''(M_i) \Rightarrow -\frac{h^2}{1r} \times rn M \leq E_{T(rn)} \leq -\frac{h^2}{1r} \times rn m \quad (1)$$

$$m = \min_{x \in [a, b]} f''(x) \quad M = \max_{x \in [a, b]} f''(x)$$

$$\Rightarrow m \leq \frac{1}{n h^2} E_{T(rn)} \leq M \Rightarrow \exists \mu \in [a, b] : f''(\mu) = \frac{1}{n h^2} E_{T(rn)}$$

طبق قضیه میانگین، در هر یک از این بازه ها، $f''(x)$ میگیرد

$$\Rightarrow E_{T(rn)} = \frac{n h^2}{1} \times f''(\mu) \quad \left\{ \Rightarrow \right.$$

$$(h = \frac{b-a}{rn})$$

طبق تعریف سوال
 $h = \frac{h}{r}$

$$E_{T(rn)} = \frac{n}{1} \times \frac{b-a}{rn} \times h^2 \times f''(\mu)$$

$$= \frac{b-a}{1r} \times \left(\frac{h}{r}\right)^2 \times f''(\mu)$$

$$E_{T(n)} = -\frac{\Lambda h'^r}{1r} \sum_{i=0}^{n-1} f''(\theta_i) \quad \xrightarrow{\text{میانگین مقدار}} \exists \mu' \in [a, b]; \quad f''(\mu') = \frac{r}{r_n h'^r} E_{T(n)} \quad (2)$$

$$h' = \frac{b-a}{r_n}$$

$$\begin{cases} -\frac{\Lambda h'^r}{1r} \times n \times M \leq E_{T(n)} \leq -\frac{\Lambda h'^r}{1r} \times n \times m \\ \Rightarrow m \leq \frac{r}{r_n h'^r} E_{T(n)} \leq M \end{cases}$$

$$E_{T(n)} = \frac{r_n h'^r}{r} f''(\mu') \quad \text{طبق (2)}$$

$$h' = \frac{b-a}{r_n}$$

$$\begin{aligned} \Rightarrow E_{T(n)} &= \frac{r_n}{r} \times \frac{b-a}{r_n} \times \left(\frac{h}{r}\right)^r \times f''(\mu') \\ &= \frac{b-a}{r} \times \left(\frac{h}{r}\right)^r \times f''(\mu') \end{aligned}$$

$$\int_a^b f(x) dx = T(n) + E_{T(n)} = T(rn) + E_{T(rn)} \quad \text{مجموعه ها را بنویسیم}$$

$$\Rightarrow T(rn) - T(n) = E_{T(n)} - E_{T(rn)}$$

$$\begin{aligned} &= -\frac{b-a}{1r} \times \left(\frac{h}{r}\right)^r \times f''(\mu) + \frac{b-a}{r} \times \left(\frac{h}{r}\right)^r \times f''(\mu') \\ &= \left(\frac{h}{r}\right)^r \times \left(\frac{b-a}{1r}\right) \left(-f''(\mu) + r f''(\mu')\right) = \left(\frac{b-a}{1r}\right) \times \left(\frac{h}{r}\right)^r \times r f''(\mu') \\ &= r E_{T(rn)} \end{aligned}$$

$$\Rightarrow \frac{T(rn) - T(n)}{r} = E_{T(rn)}$$

$$f(x) = \frac{\cos x}{x} \quad h = \frac{b-a}{n} = \frac{1.5 - 1}{r} = \frac{1}{r}$$

روش ذره‌ای:

$$\int_a^b f(x) dx \approx \frac{h}{r} \left(f(a) + f(b) + r \sum_{i=1}^{n-1} f(a + ih) \right)$$

$$\begin{aligned} \Rightarrow \int_{1/1}^{1.5} \frac{\cos x}{x} dx &= \frac{1/1}{r} \left(\frac{\cos(1/1)}{1/1} + \frac{\cos(1.5)}{1.5} + r \sum_{i=1}^r f(1/1(i+1)) \right) \\ &= \frac{1/1}{r} \left(\frac{\cos(1/1)}{1/1} + \frac{\cos(1.5)}{1.5} + r \left(\frac{\cos(1/2)}{1/2} + \frac{\cos(1/3)}{1/3} + \frac{\cos(1/4)}{1/4} \right) \right) \\ &= \frac{1/1}{r} \left(1.95 + 1.755 + r(2.9 + 1.112 + 1.103) \right) = 1.422 \end{aligned}$$

روش سیمون بکریک:

$$\int_a^b f(x) dx \approx \frac{h}{3} (f(a) + f(b) + \frac{4}{3} \sum_{i=1}^{n-1} f(a + (i+1)h) + \frac{2}{3} \sum_{i=0}^{n-1} f(a + i h))$$

$$\Rightarrow \int_{-1/1}^{1/5} \frac{\cos x}{x} dx \approx \frac{1/1}{3} \left(\frac{\cos(-1/1)}{1/1} + \frac{\cos(1/5)}{1/5} + \frac{4}{3} \sum_{i=1}^4 f(-1/1((i+1)/5)) + \frac{2}{3} \sum_{i=1}^4 f(-1/1(i/5)) \right)$$

$$= \frac{1/1}{3} \left(\frac{\cos(-1/1)}{1/1} + \frac{\cos(1/5)}{1/5} + \frac{4}{3} \left(\frac{\cos(1/2)}{1/2} + \frac{\cos(1/4)}{1/4} \right) + \frac{2 \cos(1/3)}{1/3} \right)$$

$$= \frac{1/1}{3} (9.95 + 1.755 + \frac{4}{3} (2.9 + 2.252) + 2 \times 2.182) = 1.243$$

۳- با استفاده از روش ذوزنقه:

خطا در روش ذوزنقه ای بصورت زیر می باشد:

$$|error| = \frac{n h^3}{12} f''(\mu) = \frac{(b-a) h^3}{12} f''(\mu) = \frac{(b-a)^3}{12 n^2} f''(\mu)$$

$$\mu \in [a, b]$$

یا $M = \max_{x \in [a, b]} f''(x)$ داریم:

$$|error| \leq \frac{M(b-a)^3}{12 n^2}$$

$$f(x) = \cosh(x^2) \quad f'(x) = 2x \sinh(x^2) \quad f''(x) = 2 \sinh(x^2) + 4x^2 \cosh(x^2)$$

$$\Rightarrow |error| \leq \left| \frac{1}{12 n^2} \left(\max_{x \in [0, 1]} (2 \sinh(x^2) + 4x^2 \cosh(x^2)) \right) \right| = E$$

$$E = \frac{1}{12 n^2} (2 \sinh(1) + 4 \cosh(1)) = \frac{1}{12 n^2} (2e + \frac{1}{e}) \leq 0.01$$

$$\Rightarrow \frac{2e + \frac{1}{e}}{12} \times 10^4 \leq n^2 \Rightarrow 241251 \leq n \Rightarrow$$

حد پایین مقدار مورد نیاز برای n (تعداد زیرینها) ۲۷ می باشد.
با استفاده از روش سیمون:

$$|error| \leq \left| \frac{(b-a)^5}{120 n^4} \left(\max_{x \in [a, b]} f^{(4)}(x) \right) \right|$$

$$\Rightarrow |error| \leq \frac{1}{120 n^4} \left| \max_{x \in [0, 1]} f^{(4)}(x) \right|$$

$$f^{(4)}(x) = 16x^4 \sinh(x^2) + 12x^2 \cosh(x^2) \quad (12x^2 + 12) \cosh(x^2) \quad 48x^2 \sinh(x^2)$$

$$f^{(4)}(x) = 48x^2 \sinh(x^2) + (12x^2 + 12) \cosh(x^2) \rightarrow$$

در بازه [۰، ۱] اکیدا صعودی هستند و ما کسیم آن در ۱

ادانه سوال ۳،
 $|error| \leq \frac{1}{180n^2} \left(f \sin h(1) + f \cos h(1) \right) = \frac{1}{180n^2} \left(3e - \frac{10}{e} \right)$

$$\rightarrow \frac{1}{180n^2} \left(3e - \frac{10}{e} \right) \leq 0.01 \Rightarrow \frac{10^2 \left(3e - \frac{10}{e} \right)}{180} \leq n^2$$

$$\rightarrow f_{180} \approx \sqrt{\frac{10^2 \left(3e - \frac{10}{e} \right)}{180}} \leq n$$

=> حداقل تعداد زیر بازه های مورد نیاز ۵ می باشد.

-۴
 $f(x) = \frac{x-1}{2x-x^2}$

الف) تقریب برای $f'(2.5)$ بدست آورید که خطای آن $O(h^2)$ باشد. $(h=0.1)$
 طبق سطر برای دو نقطه x_0+h و x_0-h حول نقطه x_0 داریم:

$$f(x_0+h) = f(x_0) + hf'(x_0) + \frac{h^2}{2} f''(x_0) + O(h^3)$$

$$f(x_0-h) = f(x_0) - hf'(x_0) + \frac{h^2}{2} f''(x_0) + O(h^3)$$

$$f(x_0+h) - f(x_0-h) = 2hf'(x_0) + O(h^3) \Rightarrow f'(x) = \frac{f(x_0+h) - f(x_0-h)}{2h} + O(h^2)$$

در نتیجه با استفاده از رابطه بالا داریم:

$$f'(2.5) = \frac{f(2.5+0.1) - f(2.5-0.1)}{2 \times 0.1} + O(h^2) = \frac{f(2.6) - f(2.4)}{0.2} + O(h^2)$$

$$\Rightarrow f'(2.5) = \frac{2.6-1}{2 \times 2.6 - (2.6)^2} - \frac{2.4-1}{2 \times 2.4 - (2.4)^2} + O(h^2) = 0.1048 + O(h^2)$$

ب) تقریب برای $f'(3)$ بدست آورید که خطای آن $O(h^2)$ باشد. $(h=0.1)$
 در این قسمت از سطر برای نقطه x_0+h حول x_0 استفاده می کنیم.

$$f(x_0+h) = f(x_0) + hf'(x_0) + O(h^2)$$

$$\Rightarrow f'(x_0) = \frac{f(x_0+h) - f(x_0)}{h} + O(h)$$

$$x_0 = 3, h = 0.1$$

$$\Rightarrow f'(3) = \frac{f(3+0.1) - f(3)}{0.1} + O(h) = \frac{f(3.1) - f(3)}{0.1} + O(h)$$

$$\Rightarrow f'(3) = \frac{3.1-1}{3 \times 3.1 - (3.1)^2} - \frac{3-1}{3 \times 3 - 3^2} + O(h) = \frac{0.20}{1.023} + O(h) = 0.1954 + O(h)$$

۵- مانوسن بجا تلوو برای نقاط $x_0 - 2h$, $x_0 - h$, $x_0 + h$, $x_0 + 2h$ حول نقطه x_0 بریم

$$f(x_0 - 2h) = f(x_0) - 2hf'(x_0) + 2h^2 f''(x_0) - \frac{4h^3}{6} f'''(x_0) + \frac{2h^4}{24} f^{(4)}(x_0) + O(h^5)$$

$$f(x_0 - h) = f(x_0) - hf'(x_0) + \frac{h^2}{2} f''(x_0) - \frac{h^3}{6} f'''(x_0) + \frac{h^4}{24} f^{(4)}(x_0) + O(h^5)$$

$$f(x_0 + h) = f(x_0) + hf'(x_0) + \frac{h^2}{2} f''(x_0) + \frac{h^3}{6} f'''(x_0) + \frac{h^4}{24} f^{(4)}(x_0) + O(h^5)$$

$$f(x_0 + 2h) = f(x_0) + 2hf'(x_0) + 2h^2 f''(x_0) + \frac{4h^3}{6} f'''(x_0) + \frac{2h^4}{24} f^{(4)}(x_0) + O(h^5)$$

حال با استفاده از تقصیر ضرایب $f'(x_0)$ ، $f''(x_0)$ ، $f'''(x_0)$ ، $f^{(4)}(x_0)$ در جیب $f(x_0 + 2h)$ ، $f(x_0 + h)$ ، $f(x_0 - h)$ ، $f(x_0 - 2h)$ نسبت می آوریم

$$h^3 f'''(x_0) = a f(x_0 - 2h) + b f(x_0 - h) + c f(x_0 + h) + d f(x_0 + 2h)$$

$$\Rightarrow f(x_0) : a + b + c + d = 0 \quad (1)$$

$$f'(x_0) : -2ha - hb + hc + 2hd = 0 \Rightarrow -2a - b + c + 2d = 0 \quad (2)$$

$$f''(x_0) : 2a + \frac{b}{2} + \frac{c}{2} + 2d = 0 \quad (3)$$

$$f'''(x_0) : -\frac{4}{6}a - \frac{b}{6} + \frac{c}{6} + \frac{4}{6}d = 1 \quad (4)$$

$$f^{(4)}(x_0) : \frac{2}{24}a + \frac{b}{24} + \frac{c}{24} + \frac{2}{24}d = 0 \quad (5)$$

$$(1), (2) \rightarrow b + c = 0 \Rightarrow a + d = 0$$

$$(2), (3) \rightarrow r(d - a) = b - c$$

$$\frac{4}{6}(d - a) = \frac{1}{6}(b - c) + 1 \rightarrow \left(\frac{4}{6} - \frac{1}{6}\right)(b - c) = 1$$

$$\Rightarrow b - c = 2$$

$$\Rightarrow d - a = 1$$

$$\Rightarrow \begin{cases} a = -\frac{1}{2} \\ b = 1 \\ c = -1 \\ d = \frac{1}{2} \end{cases}$$

$$\Rightarrow \frac{h^3}{h^3} f'''(x_0) = \frac{-f(x_0 - 2h) + 2f(x_0 - h) - 2f(x_0 + h) + f(x_0 + 2h)}{2} + O(h^5)$$

$$\Rightarrow f'''(x_0) = \frac{-f(x_0 - 2h) + 2f(x_0 - h) - 2f(x_0 + h) + f(x_0 + 2h)}{2h^3} + O(h^2)$$

$$I_f = \int_0^h f(x) dx \approx Qf = \frac{h}{r} (f(0) + f(h)) + \frac{h^r}{1r} (f'(0) - f'(h)) \quad \text{ـ٤}$$

$$f(x) = a_r x^r + a_{r-1} x^{r-1} + a_1 x + a_0 \quad \text{الف}$$

$$\begin{aligned} \Rightarrow \int_0^h f(x) dx &= \left. \frac{a_r}{r} x^r + \frac{a_{r-1}}{r} x^{r-1} + \frac{a_1}{r} x^r + a_0 x \right|_0^h = \frac{a_r}{r} h^r + \frac{a_{r-1}}{r} h^{r-1} + \frac{a_1}{r} h^r + a_0 h \\ &= \frac{h}{r} (a_0 + a_1 + a_{r-1} h + a_r h^{r-1} + a_r h^r) + \frac{h^r}{1r} (a_1 - a_1 - r a_{r-1} h - r a_r h^r) \\ &= \frac{h}{r} (f(h) + f(0)) + \frac{h^r}{1r} (f'(0) - f'(h)) = Q_f(h) \end{aligned}$$

$$\int_a^{a+h} f(x) dx = \int_a^h f(x+a) dx \approx \frac{h}{r} (f(a) + f(a+h)) + \frac{h^r}{1r} (f'(a) - f'(a+h)) \quad \text{ـ٥}$$

$$\Rightarrow \int_0^{nh} f(x) dx = \sum_{i=0}^{n-1} \int_{ih}^{(i+1)h} f(x) dx = \sum_{i=0}^{n-1} \int_0^h f(x+ih) dx =$$

$$\sum_{i=0}^{n-1} \left(\frac{h}{r} (f(ih) + f((i+1)h)) + \frac{h^r}{1r} (f'(ih) - f'((i+1)h)) \right)$$

$$= \frac{h}{r} (f(0) + f(nh)) + r \sum_{i=1}^{n-1} f(ih) + \frac{h^r}{1r} (f'(0) - f'(nh))$$